

**A SMART MUSHROOM INCUBATION SYSTEM FOR
ENHANCED YIELD AND SUSTAINABILITY IN SMALL-
SCALE FARMING.**

Group ID -25-26J-264

BSc (Hons) degree in Information Technology Specializing in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

April 2026

**A SMART MUSHROOM INCUBATION SYSTEM FOR
ENHANCED YIELD AND SUSTAINABILITY IN SMALL-
SCALE FARMING.**

Group ID -25-26J-264

BSc (Hons) degree in Information Technology Specializing in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

April 2026

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Signature:

Date:

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of the supervisor:

Date:

ABSTRACT

Mushroom cultivation requires precise control of environmental conditions such as temperature, humidity, carbon dioxide (CO₂), moisture, and pH levels to ensure optimal growth and productivity. Traditional farming methods rely heavily on manual monitoring and experience, which can lead to inconsistencies, delayed responses, and increased risk of contamination and crop loss.

This research presents the design and implementation of a Smart Mushroom Farming System that integrates Internet of Things (IoT) technology, machine learning, and intelligent advisory mechanisms to enhance cultivation efficiency. The system utilizes multiple sensors to collect real-time environmental data, which is transmitted through a microcontroller to a cloud-based platform for analysis. A machine learning model based on the Random Forest algorithm is employed to predict fruiting readiness and environmental trends, enabling proactive decision-making.

In addition, the system incorporates a pest and disease detection module using image-based analysis to identify contamination risks and generate real-time alerts. An intelligent advisory system further supports farmers by providing actionable recommendations for maintaining optimal growing conditions. The system is supported by a user-friendly mobile application that allows real-time monitoring, visualization, and control.

The results demonstrate that the proposed system improves accuracy, reduces manual effort, and enhances productivity by enabling data-driven decision-making. The integration of monitoring, prediction, detection, and advisory components provides a comprehensive solution for smart agriculture. This system has strong potential for real-world application, particularly for small and medium-scale farmers seeking efficient and sustainable farming practices.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who supported and guided us throughout the successful completion of this project.

First and foremost, we would like to thank our project supervisor for their valuable guidance, continuous support, and insightful feedback, which greatly contributed to the development of this research. Their expertise and encouragement helped us to overcome challenges and improve the quality of our work.

We also extend our appreciation to the academic staff and the institution for providing the necessary resources, facilities, and knowledge that enabled us to carry out this project effectively. Their support created a strong foundation for our learning and research activities.

We would like to acknowledge the contributions of our group members for their teamwork, collaboration, and dedication. Each member played an important role in completing different components of the system, and the project would not have been successful without their collective effort.

Finally, we are grateful to our families and friends for their constant encouragement, patience, and support throughout the project duration.

This project would not have been possible without the contribution of all these individuals and organizations.

TABLE OF CONTENTS

DECLARATION	iii
ABSTRACT	iv
ACKNOWLEDGEMENT	v
LIST OF FIGURES.....	viii
LIST OF TABLES.....	ix
LIST OF ABBREVIATIONS.....	x
INTRODUCTION	1
Background and Literature Survey	2
Research Gap	5
Research Problem.....	7
Research Objectives	8
Main Objectives	8
Specific Objectives.....	9
METHODOLOGY	10
Methodology	10
System Overview	10
Data Collection and Monitoring Module	11
Prediction Module	13
Fruiting Prediction Module	15
Pest and Disease Detection Module	17
System Architecture	18
System Requirements	19
Commercialization Aspects of the Product	20
Target Market and Business Model.....	22
Product Components and Packaging.....	23
Revenue Model and Pricing Strategy.....	24

Scalability and Deployment Strategy.....	25
Challenges and Commercial Risks	26
Testing and Implementation.....	26
System Implementation.....	26
Sensors Testing and Calibration.....	27
Communication and Data Transmission Testing.....	28
Machine Learning Model Testing	29
System Integration Testing.....	29
User Interface Testing	30
Performance Evaluation	30
Limitations Identified During Testing.....	30
RESULTS AND DISCUSSION.....	31
Results	31
Research Findings	40
Discussion	41
3.4 Summary of Each Student’s Contribution	43
CONCLUSION.....	44
REFERENCES.....	47
GLOSSARY	49
APPENDICES	50

LIST OF FIGURES

- Figure 1.1 Factors Affecting Mushroom Growth
- Figure 1.2 IoT-Based Smart Agriculture System
- Figure 1.3 Research Gap in Existing Systems

- Figure 2.1 System Workflow Overview
- Figure 2.2 Data Collection and Transmission Process
- Figure 2.3 Prediction Model Workflow
- Figure 2.4 Prediction Model Graph
- Figure 2.5 Fruiting Prediction Process
- Figure 2.6 Pest and Disease Detection Workflow
- Figure 2.7 System Architecture
- Figure 2.8 Commercial Workflow and Business Model
- Figure 2.9 System Implementation Workflow
- Figure 2.10 Actual vs Predicted Values Comparison

- Figure 3.1 Sensor Monitoring and Control Interface
- Figure 3.2 Real-Time Dashboard and System Status
- Figure 3.3 Environmental Parameter Trends Over Time
- Figure 3.4 Comparison Between Actual and Predicted Values
- Figure 3.5 Pest and Disease Detection Alerts
- Figure 3.6 AI Advisory System Interface
- Figure 3.7 Robotic Control Interface
- Figure 3.8 Machine Learning Model Interface

LIST OF TABLES

Table 1.1	Comparison of Existing Systems
Table 1.2	Research Gap Summary
Table 2.1	Sensor Parameters and Measurement Details
Table 2.2	System Requirements
Table 2.3	Summary of System Components
Table 2.4	Proposed Pricing Model
Table 2.5	Sensor Accuracy Levels and Performance
Table 3.1	Performance Metrics (MAE, RMSE, Accuracy)

LIST OF ABBREVIATIONS

IoT	Internet of Things
ML	Machine Learning
AI	Artificial Intelligence
CNN	Convolutional Neural Network
CO ₂	Carbon Dioxide
MQTT	Message Queuing Telemetry Transport
HTTP	Hypertext Transfer Protocol
ESP32	Microcontroller Unit for IoT Applications
RF	Random Forest
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
UI	User Interface
API	Application Programming Interface
DHT22	Temperature and Humidity Sensor
YOLO	You Only Look Once (Object Detection Algorithm)

INTRODUCTION

Mushroom cultivation has emerged as a rapidly growing agricultural sector due to its high nutritional value, medicinal properties, and economic potential. In countries such as Sri Lanka, it plays an important role in supporting small-scale farmers by providing a sustainable source of income with relatively low initial investment requirements. Among different varieties, oyster mushrooms are widely cultivated because of their fast growth cycle and adaptability to local environmental conditions. However, successful cultivation depends heavily on maintaining precise environmental and substrate conditions throughout the growth cycle.

Mushrooms are highly sensitive to variations in environmental parameters such as temperature, humidity, carbon dioxide (CO₂) concentration, substrate moisture, and pH levels. Even slight deviations from optimal conditions can delay fruiting, reduce yield quality, and increase the risk of contamination. Traditionally, farmers rely on manual observation and experience to monitor and control these parameters, which is often time-consuming, inconsistent, and prone to human error. As a result, many small-scale cultivators struggle to achieve consistent production and optimal resource utilization.

Recent advancements in smart agriculture technologies have introduced new opportunities to overcome these challenges. The integration of Internet of Things (IoT) systems enables real-time monitoring of environmental conditions through sensor networks, allowing farmers to collect continuous and accurate data [1], [2]. In addition, machine learning techniques have been widely adopted to analyze agricultural data and provide predictive insights, improving decision-making processes [3], [12]. Furthermore, computer vision and deep learning approaches have shown significant success in detecting pests and diseases in crops, reducing dependency on manual inspection [7], [37].

Despite these advancements, most existing solutions focus on individual functionalities such as monitoring, prediction, or detection, rather than providing a

fully integrated system. This limitation highlights the need for a comprehensive solution that combines multiple technologies into a single platform. Therefore, this research presents an integrated Smart Mushroom Farming System that incorporates IoT-based monitoring, machine learning-based prediction, computer vision-based detection, and an intelligent advisory system to improve efficiency, accuracy, and productivity in mushroom cultivation.

Background and Literature Survey

Mushroom cultivation is highly dependent on environmental control, where factors such as temperature, humidity, and carbon dioxide levels directly influence growth rate and yield quality. Previous studies have emphasized the importance of maintaining stable environmental conditions to ensure optimal mushroom development [4], [10]. In traditional farming practices, these conditions are managed manually, which often leads to inconsistencies and inefficiencies.

As illustrated in Figure 1.1, multiple environmental factors influence mushroom growth, highlighting the need for precise monitoring and control.

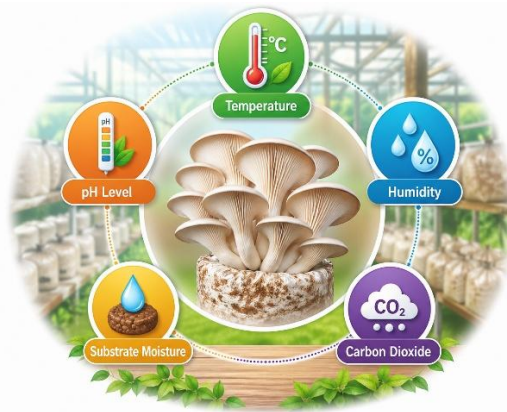


Figure 1.1: Factors affecting mushroom growth

With the evolution of smart agriculture, IoT-based systems have been widely adopted to automate environmental monitoring. Wireless sensor networks enable continuous data collection from the cultivation environment, allowing real-time tracking of key

parameters [2], [23]. These systems improve accuracy and reduce manual labor; however, many of them are limited to monitoring and lack predictive capabilities.

As illustrated in Figure 1.2, IoT-based systems provide a structured framework for real-time data collection, transmission, and analysis in smart agriculture.

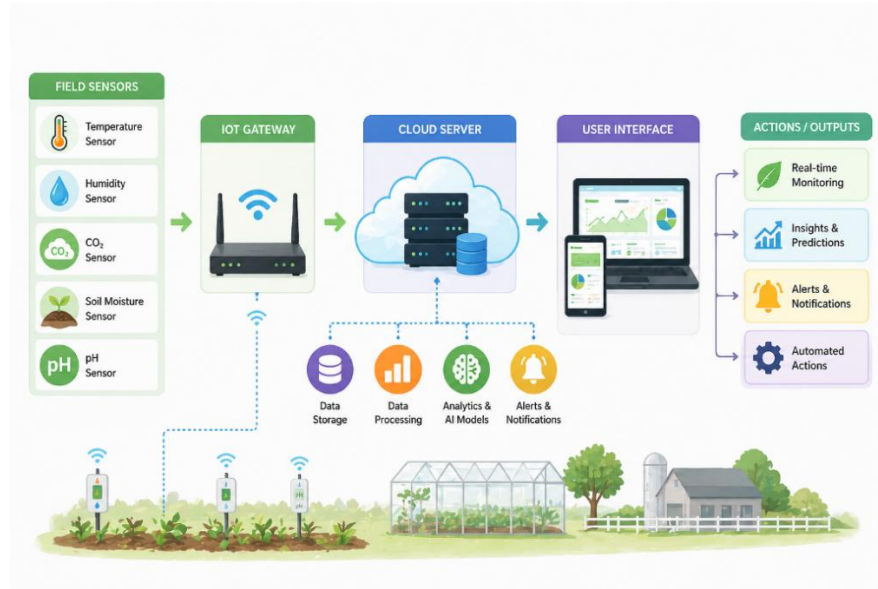


Figure 1.2: IoT-based smart agriculture system

Machine learning has further enhanced agricultural systems by enabling predictive analysis based on historical and real-time data. Techniques such as time-series forecasting and deep learning models have been successfully used to predict environmental changes and crop behavior [17], [46]. In mushroom cultivation, predictive models can help estimate fruiting readiness and optimize harvesting schedules, reducing uncertainty for farmers.

In addition, computer vision technologies have been widely applied in agriculture for pest and disease detection. Convolutional Neural Networks (CNNs) and object detection models such as YOLO have demonstrated high accuracy in identifying pests and plant diseases from images [14], [37]. These approaches reduce the need for manual inspection and enable faster response to potential threats.

However, a key limitation in existing research is the lack of integration between these technologies. Most systems operate independently, focusing either on monitoring, prediction, or detection. This fragmented approach reduces overall efficiency and limits the ability to provide comprehensive decision support.

A comparison of existing systems and their limitations is presented in Table 1.1.

N o	Study / System	Real- time Monitor ing	Predicti on Capabil ity	Pest / Diseas e Detecti on	Adviso ry Suppo rt	Focus on Mushro om Cultivat ion	Key Limitatio n
1	IoT-based Monitorin g Systems	✓	✗	✗	✗	✗	Only monitors environme ntal parameters
2	Smart Agricultur e IoT Systems	✓	✗	✗	✗	✗	General agriculture , not mushroom -specific
3	Machine Learning Prediction Models	✗	✓	✗	✗	✗	Not integrated with real- time data
4	Crop Yield Prediction Systems	✓	✓	✗	✗	✗	Focus on crops, not mushroom substrate

5	Pest Detection Systems (CNN/YOLO)	✗	✗	✓	✗	✗	No prediction or advisory integration
6	Precision Agriculture Decision Systems	✓	✓	✗	✓	✗	No focus on pH or mushroom conditions
7	Proposed System	✓	✓	✓	✓	✓	Fully integrated smart system

Table 1.1: Comparison of Existing Systems

Research Gap

Although significant advancements have been made in applying IoT, machine learning, and computer vision in agriculture, several critical limitations remain when these technologies are applied to mushroom cultivation. Existing IoT-based systems primarily focus on real-time environmental monitoring but lack predictive capabilities, forcing farmers to rely on reactive decision-making [11], [23]. Similarly, machine learning models are often developed independently and are not integrated with real-time monitoring systems, reducing their practical usability [12], [50].

Computer vision-based systems for pest detection also operate as standalone solutions without providing advisory or automated response mechanisms [7], [30]. Furthermore, many existing solutions are designed for large-scale operations and are not suitable for small-scale farmers due to high costs and complexity.

As illustrated in Figure 1.3, existing systems lack integration between monitoring, prediction, and advisory functionalities, highlighting the research gap addressed in this study.

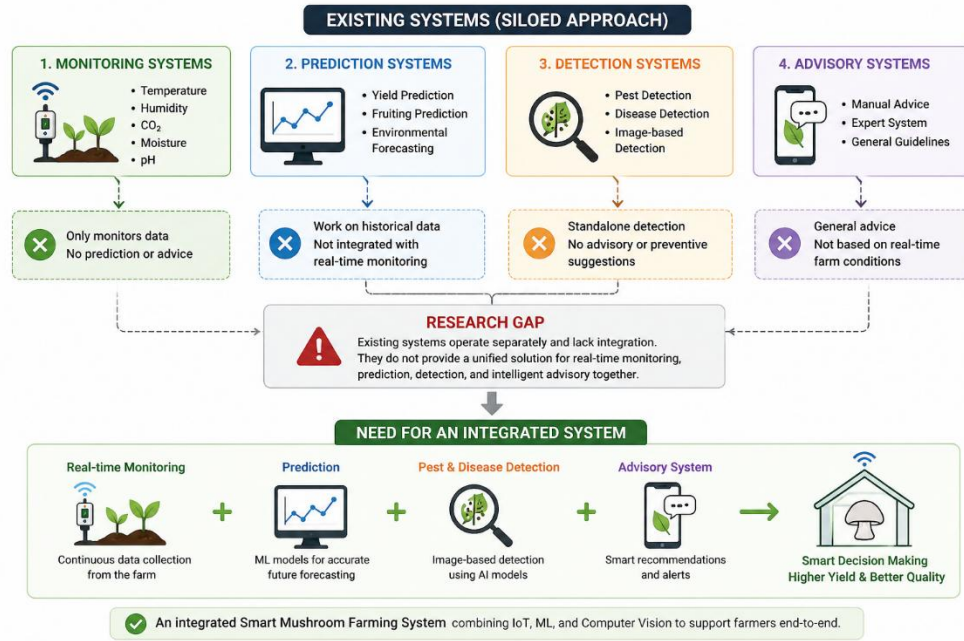


Figure 1.3: Research gap in existing systems

A structured summary of the identified research gaps and the proposed system capabilities is presented in Table 1.2.

No	Feature / Capability	Existing Systems	Proposed System
1	Real-time Environmental Monitoring	✓ Available	✓ Available
2	Prediction of Moisture & pH	✗ Not available or limited	✓ Implemented
3	Fruiting Prediction	✗ Not available	✓ Implemented
4	Pest / Disease Detection	✓ Available (standalone)	✓ Integrated

5	Advisory / Decision Support	✗ Not available	✓ Implemented
6	Integration of All Modules	✗ Separate systems	✓ Fully integrated
7	Real-time Data + Prediction Combination	✗ Not integrated	✓ Integrated
8	Suitability for Small-scale Farmers	✗ Limited / expensive	✓ Cost- effective solution

Table 1.2: Research Gap Summary

The table clearly highlights the limitations of existing systems and justifies the need for the proposed integrated solution.

Research Problem

Mushroom cultivation is a highly sensitive agricultural process that requires precise control of environmental and substrate conditions to ensure optimal growth and yield. In small-scale farming environments, particularly in Sri Lanka, cultivators face significant challenges in maintaining these conditions due to the reliance on manual monitoring and experience-based decision-making. Environmental parameters such as temperature, humidity, carbon dioxide concentration, substrate moisture, and pH levels fluctuate continuously, and even minor deviations can negatively impact mushroom growth, delay fruiting, and increase the risk of contamination.

One of the major issues is the inability of farmers to monitor multiple parameters simultaneously with high accuracy. Manual observation is time-consuming and often fails to detect subtle changes in environmental conditions. This leads to delayed responses, resulting in reduced productivity and inconsistent yield quality.

Additionally, pest infestations and fungal contamination can spread rapidly if not detected early, causing significant losses.

Existing technological solutions provide partial support but fail to address the problem comprehensively. IoT-based systems offer real-time monitoring but lack predictive capabilities, while machine learning models provide predictions without being fully integrated into real-time data environments. Similarly, pest detection systems using computer vision focus only on identification and do not incorporate automated response mechanisms or decision support features.

Furthermore, many available systems are designed for large-scale industrial farms and are not suitable for small-scale farmers due to high costs and technical complexity. This creates a gap between technological advancements and practical usability in real-world farming conditions.

Therefore, the central research problem addressed in this study is how to design and implement an integrated, affordable, and intelligent system that can continuously monitor environmental conditions, predict future changes, detect pests and diseases, and provide actionable recommendations to farmers. Solving this problem is essential to improving efficiency, reducing manual effort, and enhancing the sustainability and productivity of mushroom cultivation.

Research Objectives

Main Objectives

The main objective of this research is to design, develop, and evaluate an integrated Smart Mushroom Farming System that leverages Internet of Things (IoT), machine learning, and computer vision technologies to improve real-time monitoring, predictive analysis, pest detection, and decision-making processes in mushroom cultivation. The system aims to enhance productivity, reduce manual effort, and provide a cost-effective and scalable solution for small-scale farmers.

Specific Objectives

- To design and implement an IoT-based monitoring system capable of collecting real-time environmental data, including temperature, humidity, and carbon dioxide levels, within the mushroom cultivation environment.
- To develop machine learning models for predicting substrate moisture levels, pH variations, and fruiting readiness based on historical and real-time data, enabling proactive decision-making.
- To design and implement a computer vision-based detection system using deep learning techniques to identify pest activity and potential contamination within the mushroom farm.
- To develop an intelligent advisory system that integrates outputs from monitoring, prediction, and detection modules to generate real-time alerts and actionable recommendations for farmers.
- To evaluate the performance of the proposed system using relevant metrics such as prediction accuracy, detection accuracy, system reliability, and usability in real-world conditions.
- To ensure that the system is cost-effective, scalable, and suitable for adoption by small-scale mushroom farmers, particularly in developing regions.

METHODOLOGY

Methodology

System Overview

The proposed Smart Mushroom Farming System was designed and implemented as an integrated solution that combines Internet of Things (IoT), machine learning, and computer vision technologies to address the limitations of traditional mushroom cultivation practices. The system was structured into four primary modules: Monitoring Module, Prediction Module, Fruiting Prediction Module, and Pest/Disease Detection Module. These modules were interconnected through a centralized architecture to ensure continuous data flow and coordinated system behavior.

The Monitoring Module continuously collected environmental data such as temperature, humidity, carbon dioxide (CO₂) levels, and substrate moisture using IoT sensors. This real-time data served as the foundation for all subsequent analysis and decision-making processes within the system. The Prediction Module processed this data using machine learning algorithms to forecast key parameters such as moisture levels and pH variations. The Fruiting Prediction Module further analyzed environmental trends and growth patterns to determine optimal harvesting times. The Pest and Disease Detection Module utilized image processing and deep learning techniques to identify contamination and pest activity within the cultivation environment.

The integration of these modules enabled the system to operate as a unified intelligent platform, where outputs from one module directly influenced the behavior of others. For example, sensor data collected by the Monitoring Module was used as input for predictive models, while detection results triggered advisory recommendations. This interconnected design ensured that the system not only monitored conditions but also predicted, detected, and responded to changes in real time.

The overall workflow of the system, including data flow from sensors to prediction and advisory components, is illustrated in Figure 2.1.

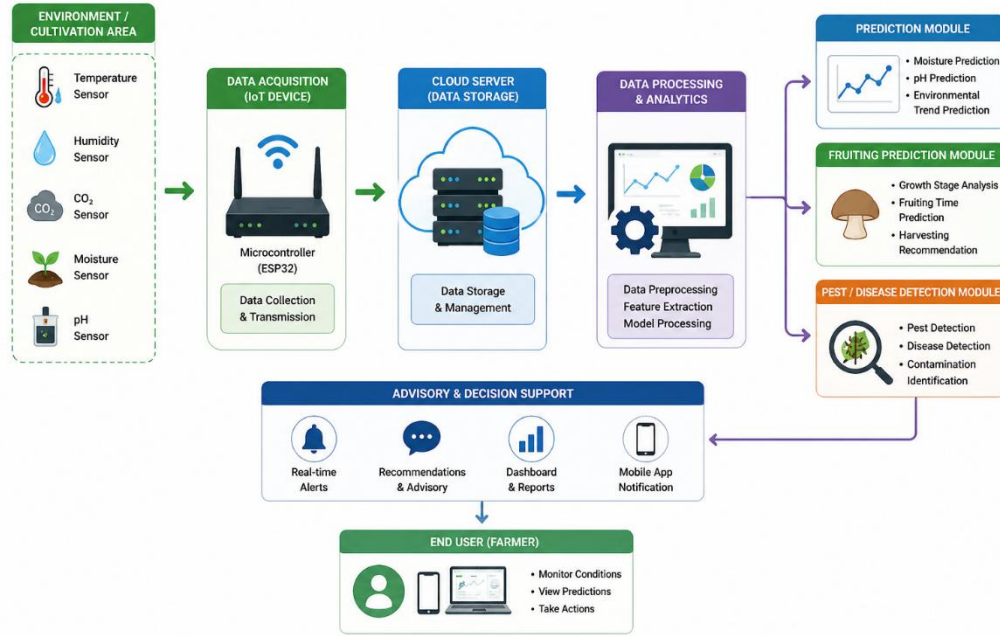


Figure 2.1: System workflow overview

Data Collection and Monitoring Module

The data collection process was carried out using IoT-based sensing devices deployed within the mushroom cultivation environment. The primary objective of this module was to continuously monitor environmental conditions that directly influence mushroom growth. Sensors were used to measure temperature, relative humidity, carbon dioxide concentration, and substrate moisture. In addition, pH levels were periodically measured to ensure that the substrate remained within the optimal range required for mushroom development.

The sensors were connected to a microcontroller unit, such as ESP32, which was responsible for collecting and transmitting data. The microcontroller processed raw sensor readings and sent them to a cloud-based storage system using wireless communication protocols such as Wi-Fi. This enabled real-time data availability and remote monitoring capabilities.

Data were collected at regular intervals and stored in a structured format for further processing. Preprocessing steps were applied to ensure data quality, including handling missing values, filtering noise, and normalizing data ranges. These steps were essential to improve the accuracy of predictive models and ensure reliable system performance.

Compared to manual monitoring methods, the IoT-based approach significantly improved data accuracy, reduced human effort, and enabled continuous observation of environmental changes. This allowed the system to detect even minor fluctuations in conditions, which is critical for maintaining optimal growth environments.

The process of collecting, transmitting, and storing environmental data is illustrated in Figure 2.2, which shows how sensor readings are sent from the cultivation environment to the cloud system.

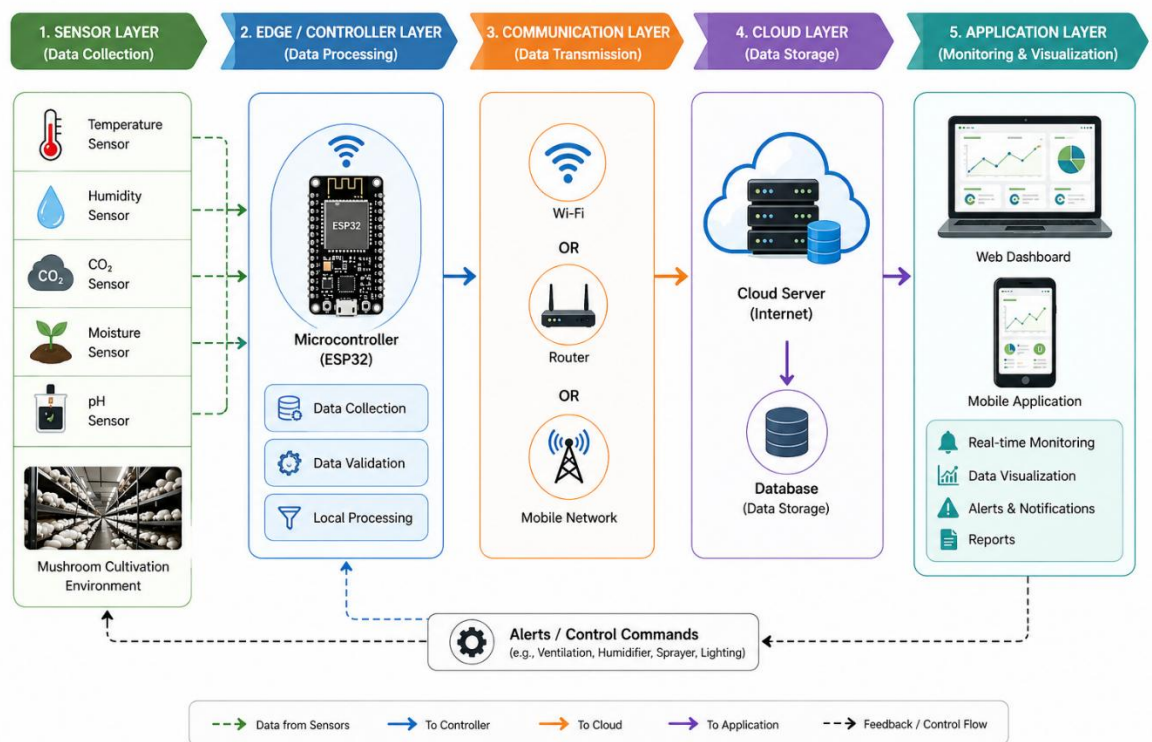


Figure 2.2: Data collection and transmission process

Furthermore, the details of the sensors and their measurement purposes are summarized in Table 2.1.

Parameter	Sensor Type	Purpose
Temperature	DHT22	Monitor growth temperature
Humidity	DHT22	Maintain humidity level
CO ₂	MQ Sensor	Air quality monitoring
Moisture	Soil Sensor	Substrate condition
pH	pH Sensor	Measure acidity

Table 2.1: Sensor Parameters and Measurement Details

Prediction Module

The Prediction Module was developed to provide forecasting capabilities based on environmental and historical data. The primary goal of this module was to predict substrate moisture levels, pH variations, and environmental trends to enable proactive decision-making.

Time-series forecasting techniques were used to model sequential data collected from sensors. Among these, Random Forest algorithm was selected due to its ability to handle multi-dimensional environmental data and provide accurate predictions. The dataset was divided into training and testing sets, and preprocessing techniques such as normalization and scaling were applied to improve model performance.

The model was trained using historical sensor data, where patterns in environmental changes were learned over time. After training, the model was deployed to generate real-time predictions based on incoming data. These predictions were used to identify potential deviations from optimal conditions before they occurred.

The implementation of predictive modeling significantly enhanced the system's ability to respond proactively. Instead of reacting to unfavorable conditions after they occur,

the system was able to anticipate changes and provide early recommendations, improving overall efficiency and yield consistency.

The overall prediction process, including data preprocessing, training, and prediction stages, is illustrated in Figure 2.3.

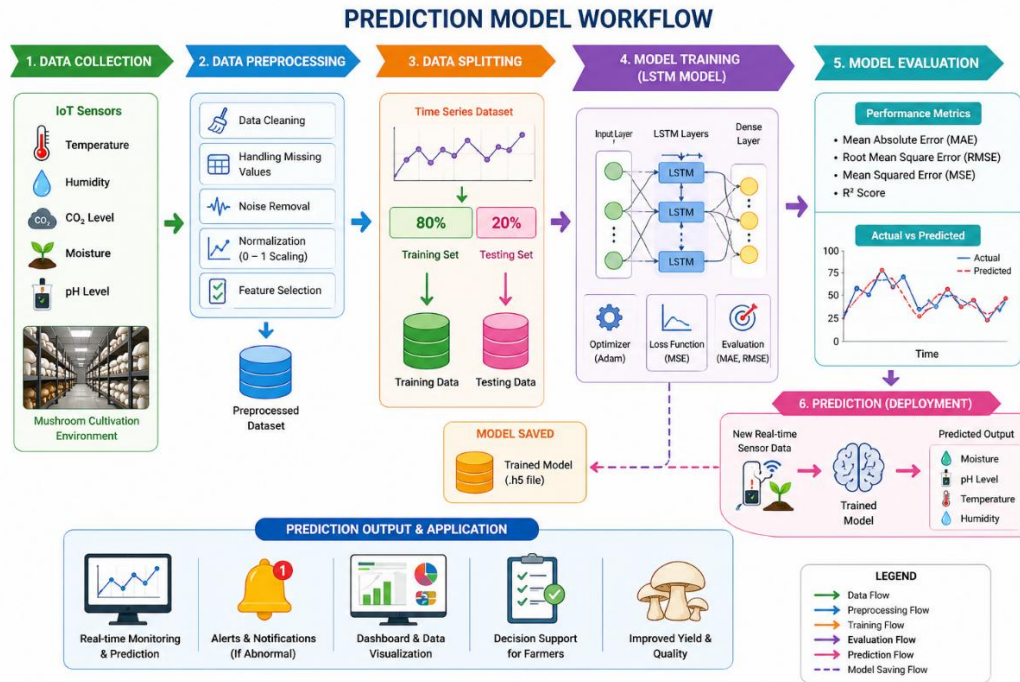


Figure 2.3: Prediction Model Workflow

The performance of the prediction model was evaluated by comparing actual and predicted values, as shown in the graph presented in Figure 2.4. This comparison demonstrates the accuracy and reliability of the model in predicting environmental conditions.



Figure 2.4: Prediction Model Graph

Fruiting Prediction Module

The Fruiting Prediction Module was developed to estimate the optimal harvesting time based on environmental conditions and growth patterns. This module analyzed trends in temperature, humidity, and substrate conditions to determine the stages of mushroom growth.

Machine learning techniques were used to classify growth stages, including incubation, pinhead formation, and fruiting stages. By identifying patterns associated with each stage, the system was able to predict when mushrooms would reach maturity.

This prediction allowed farmers to optimize harvesting schedules, reduce premature harvesting, and improve product quality. The module also contributed to better resource management by aligning environmental conditions with growth requirements.

The integration of fruiting prediction with other modules ensured that environmental control, prediction, and harvesting decisions were aligned, resulting in a more efficient cultivation process.

The stages of mushroom growth and the prediction process are illustrated in Figure 2.5.



Figure 2.5: Fruiting prediction process

Pest and Disease Detection Module

The Pest and Disease Detection Module was implemented using computer vision and deep learning techniques to identify contamination and pest activity in the mushroom cultivation environment. Images were captured using cameras positioned within the cultivation area, ensuring coverage of all critical zones.

Convolutional Neural Networks (CNNs) and object detection models such as YOLO were used for image analysis. The model was trained using labeled datasets containing images of healthy and infected mushrooms. Preprocessing steps such as resizing, normalization, and augmentation were applied to improve model accuracy and robustness.

The trained model was capable of detecting pests and contamination in real time. When abnormalities were identified, the system generated alerts and triggered advisory recommendations. This enabled early intervention, reducing crop loss and improving overall productivity.

The use of computer vision significantly reduced the need for manual inspection and allowed continuous monitoring of the cultivation environment, enhancing system efficiency.

The workflow of the detection process, including image input, processing, and output, is illustrated in Figure 2.6.

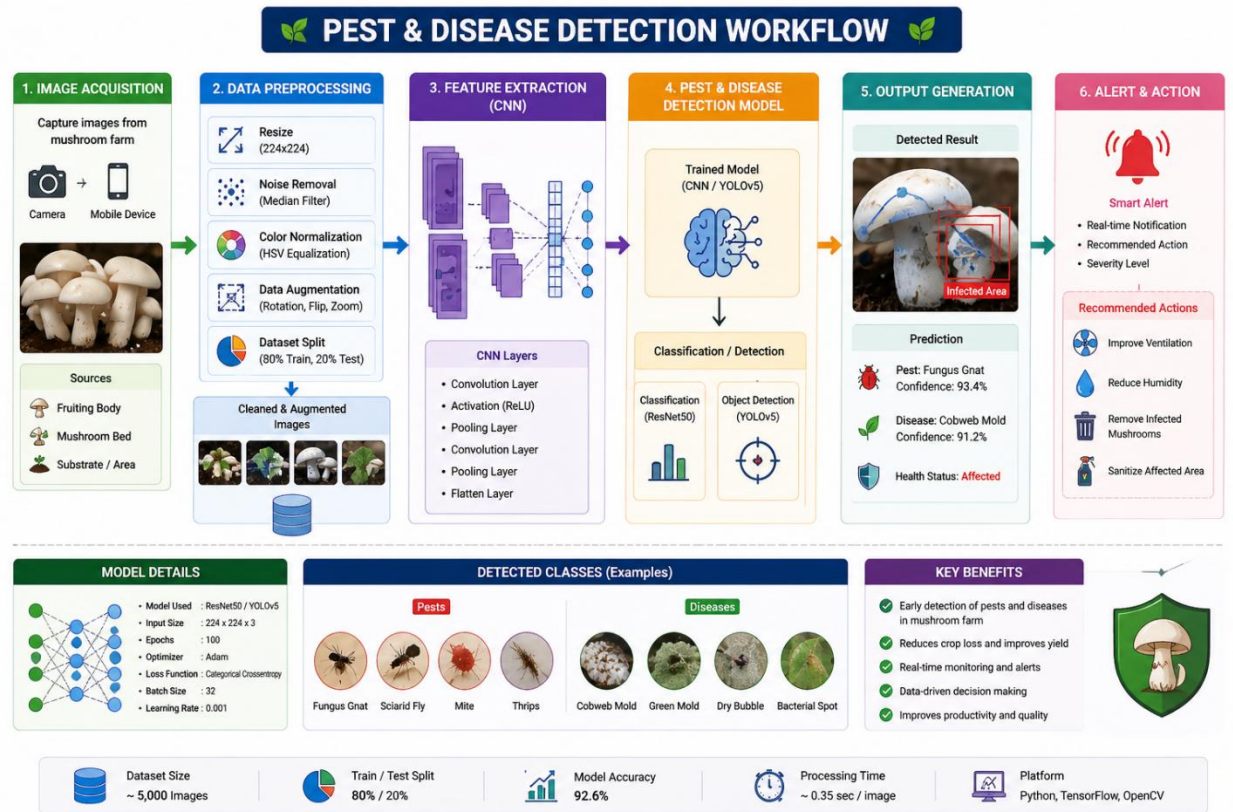


Figure 2.6: Pest and disease detection workflow

System Architecture

The overall system architecture was designed using a layered approach to ensure scalability and efficient data flow. The architecture consists of four main layers: sensing layer, communication layer, processing layer, and application layer.

The sensing layer includes IoT sensors responsible for collecting environmental data. The communication layer handles data transmission using wireless protocols. The processing layer performs data analysis, prediction, and detection using machine learning models. The application layer provides a user interface for monitoring and decision-making.

This layered architecture ensures that each component operates efficiently while maintaining seamless integration between modules.

The complete system architecture is illustrated in Figure 2.7.

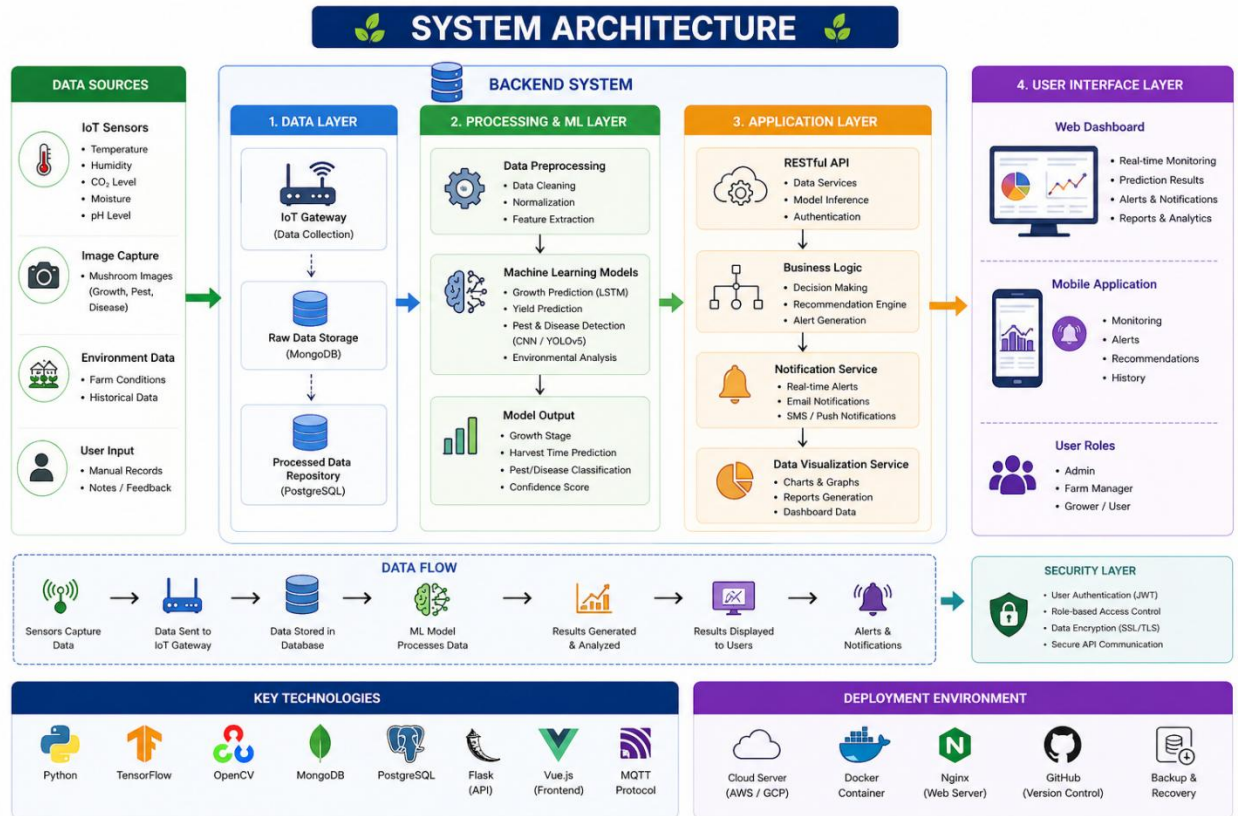


Figure 2.7: System Architecture

System Requirements

Requirement Type	Description
Functional	Monitoring, prediction, detection
Non-functional	Reliability, scalability
Hardware	Sensors, controller
Software	ML models, dashboard

Table 2.2: System Requirements

Commercialization Aspects of the Product

The proposed Smart Mushroom Farming System demonstrates strong potential for commercialization due to its practical relevance, scalability, and ability to address real-world challenges in modern agriculture. The system is specifically designed to support mushroom farmers by improving productivity, reducing crop loss, and enabling data-driven decision-making.

One of the key strengths of the system is its ability to operate as an integrated smart agriculture solution that combines IoT-based monitoring, machine learning prediction, pest detection, and advisory services within a single platform. This integration reduces the need for multiple standalone systems and simplifies farm management, making it highly attractive for commercial deployment.

From a market perspective, the system targets small-scale and medium-scale mushroom farmers, particularly in developing countries such as Sri Lanka, where access to advanced agricultural technologies is limited. Mushroom cultivation is a growing industry in Sri Lanka due to its relatively low space requirements and high nutritional value. However, farmers often face challenges related to environmental control, contamination, and lack of technical knowledge. The proposed system directly addresses these challenges, making it a viable commercial product.

The system can be commercialized as a bundled solution consisting of hardware, software, and support services. The hardware component includes IoT sensors such as temperature, humidity, CO₂, moisture, and pH sensors, along with a microcontroller (ESP32) for data processing and communication. The software component includes the mobile application, cloud-based data storage, and machine learning models for prediction and advisory. These components can be packaged as a complete “Smart Farming Kit” for end users.

In terms of pricing strategy, the system can be designed to be cost-effective and affordable for farmers. A tiered pricing model can be adopted, where basic monitoring features are offered at a lower cost, while advanced features such as prediction,

advisory, and automation are available through subscription-based services. This approach ensures accessibility while also generating continuous revenue.

The scalability of the system is another important factor supporting commercialization. The modular architecture allows additional sensors, modules, or features to be integrated without redesigning the entire system. This enables farmers to expand their operations gradually while continuing to use the same platform.

Furthermore, the system supports remote monitoring and cloud integration, allowing farmers to access data and receive recommendations from anywhere. This is particularly beneficial for large-scale farms or farmers managing multiple cultivation units. The use of mobile technology ensures ease of use and increases adoption among users.

Potential revenue streams for commercialization include direct sales of hardware kits, subscription fees for advisory and prediction services, maintenance and technical support services, and data-driven agricultural insights. The system can also be integrated with government agricultural programs, research institutions, and private sector partnerships to promote wider adoption.

In addition, the system aligns with global trends in precision agriculture and sustainable farming. By optimizing environmental conditions and reducing resource wastage, the system contributes to environmentally sustainable practices. This increases its value in both local and international markets.

Despite its strong commercialization potential, certain challenges must be considered. Initial setup costs, user training requirements, and technical maintenance may act as barriers for some farmers. Therefore, providing user training, technical support, and simplified installation processes will be essential for successful adoption.

Overall, the proposed Smart Mushroom Farming System presents a highly feasible and scalable commercial product. Its ability to combine affordability, usability, and advanced technology positions it as a competitive solution in the emerging smart agriculture market.

Target Market and Business Model

The target market for the proposed Smart Mushroom Farming System primarily includes small-scale and medium-scale mushroom farmers, agricultural entrepreneurs, and greenhouse operators. In countries such as Sri Lanka, mushroom cultivation is increasingly popular due to its low land requirements and high economic return. However, many farmers lack access to advanced technologies for environmental monitoring and decision-making.

The system is designed to be affordable and user-friendly, making it suitable for farmers with limited technical expertise. In addition, agricultural research institutions and training centers can also adopt the system for educational and experimental purposes.

From a business model perspective, the system can be commercialized using a hybrid approach that combines hardware sales with subscription-based services. Farmers can purchase the initial hardware kit, which includes sensors and a controller, while accessing advanced features such as predictive analytics and advisory recommendations through a subscription model.

The overall business workflow of the system is illustrated in Figure 2.8, which shows how the product is delivered, used, and monetized within the agricultural ecosystem.

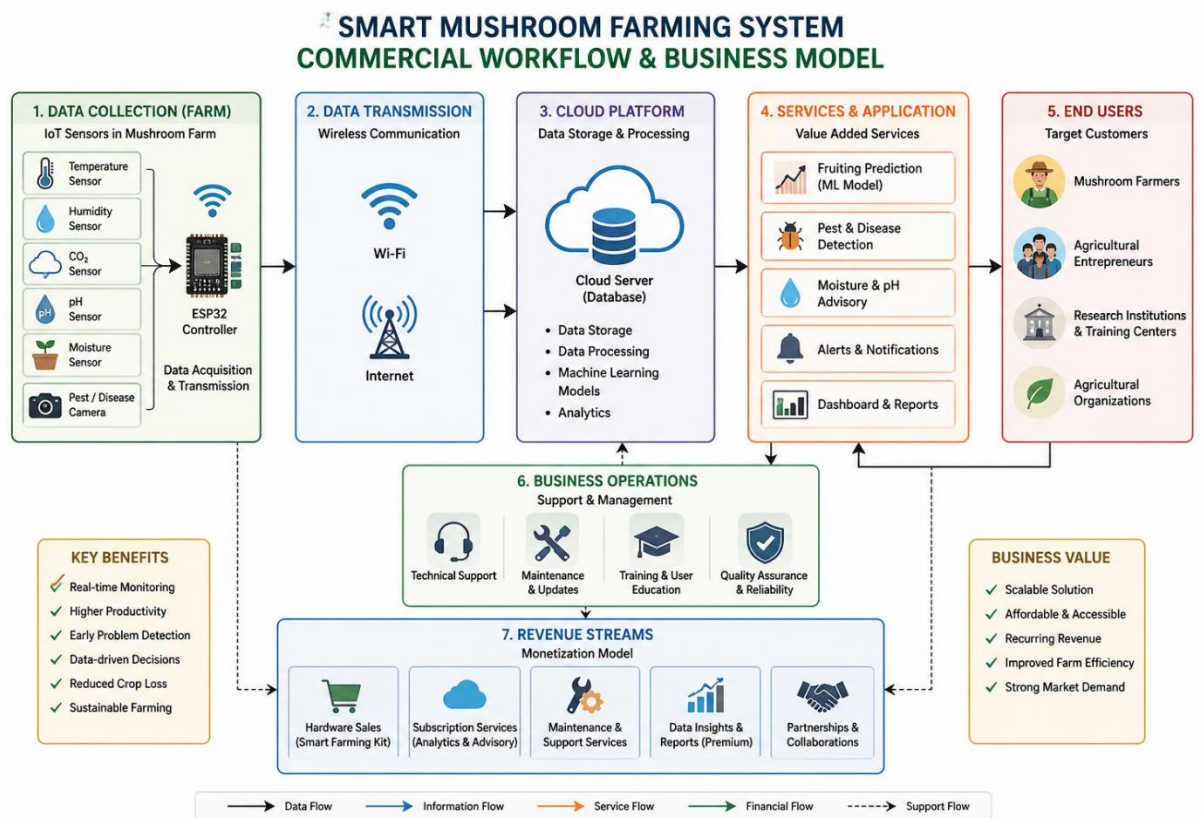


Figure 2.8: Commercial workflow and business model of the smart farming system

Product Components and Packaging

The Smart Mushroom Farming System can be commercialized as a complete product package consisting of hardware, software, and service components. The hardware package includes environmental sensors, microcontroller units, and connectivity modules required for data collection and transmission. The software package includes the mobile application, cloud platform, and machine learning models responsible for prediction and advisory functionalities.

The product can be delivered as a ready-to-install kit, allowing farmers to easily set up the system without requiring advanced technical knowledge. Additional services such as installation support, maintenance, and user training can further enhance product value.

Table 2.3 presents the key components of the system and their respective roles within the overall solution.

Component	Technology Used	Function
Sensors	DHT22, MQ, pH Sensor	Collect real-time environmental data
Microcontroller	ESP32	Process data and manage communication
Communication	Wi-Fi (MQTT/HTTP)	Transmit data to cloud system
Machine Learning	Random Forest Algorithm	Predict fruiting readiness and conditions
Mobile Application	Android Application	Provide user interface for monitoring/control
Cloud Platform	Firebase / Backend	Store, process, and analyze data
Detection Module	CNN / YOLO	Detect pests and contamination
Advisory System	AI-based Decision Logic	Provide recommendations to users

Table 2.3: Summarize of system Components

Table 2.3 summarizes the key components of the proposed system along with the technologies used and their respective functions within the overall architecture.

Revenue Model and Pricing Strategy

The commercialization of the system can be supported through multiple revenue streams. The primary revenue source is the sale of the hardware kit, which provides the core functionality of the system. In addition, subscription-based services can be introduced for advanced features such as predictive analytics, AI advisory, and automated alerts.

A tiered pricing strategy can be adopted to accommodate different categories of users. Basic users can access essential monitoring features at a lower cost, while premium users can benefit from advanced capabilities such as machine learning predictions and detailed insights.

Table 2.4 outlines a possible pricing structure for the system, demonstrating how different service levels can be offered to users.

Package Type	Features Included	Pricing Model
Basic	Sensor monitoring and dashboard access	One-time purchase
Standard	Monitoring with alerts and notifications	Monthly subscription
Premium	Full system including ML prediction, advisory, and automation	Subscription + setup

Table 2.4: Proposed Pricing Model

Table 2.4 presents the proposed pricing strategy for the system, highlighting different service levels designed to meet the needs of various users while ensuring affordability and scalability.

Scalability and Deployment Strategy

The proposed system is designed with a modular architecture, allowing it to scale according to user requirements. Farmers can start with a basic setup and gradually expand by adding more sensors or modules. This flexibility ensures that the system remains cost-effective and adaptable to different farming scales.

Deployment can be carried out in phases, beginning with pilot testing in selected farms, followed by gradual expansion to larger agricultural communities. Partnerships with government agencies, agricultural organizations, and private sector stakeholders can further support large-scale deployment.

The scalability of the system ensures that it can be adapted not only for mushroom cultivation but also for other types of controlled-environment agriculture.

Challenges and **Commercial Risks**

Despite its strong commercialization potential, several challenges must be considered. Initial setup costs may be a barrier for small-scale farmers, and proper training is required to ensure effective use of the system. Additionally, system performance depends on reliable internet connectivity, which may not be consistently available in rural areas.

Technical maintenance, sensor calibration, and long-term system reliability are also important factors that can influence user adoption. Addressing these challenges through user support, training programs, and affordable pricing strategies will be essential for successful commercialization.

Testing and Implementation

The Smart Mushroom Farming System was implemented and evaluated to ensure its reliability, accuracy, and real-time performance under practical conditions. The testing and implementation process involved integrating hardware components, software modules, and communication systems into a unified platform and validating their functionality through systematic testing procedures.

System Implementation

The system implementation was carried out using a modular approach, allowing each component to be developed and tested independently before full system integration. The hardware setup consisted of IoT sensors including temperature and humidity sensors (DHT22), CO₂ sensors (MQ series), soil moisture sensors, and pH sensors.

These sensors were connected to an ESP32 microcontroller, which served as the central processing unit for data acquisition and communication.

The ESP32 collected sensor readings at regular intervals and transmitted the data to a cloud-based backend system using Wi-Fi communication. The backend system handled data storage, preprocessing, and integration with machine learning models. The mobile application was developed to provide a user interface for real-time monitoring, visualization, and control.

The overall implementation architecture ensured continuous data flow from sensors to the cloud and then to the user interface. The interaction between system components is illustrated in Figure 2.9, which shows the complete implementation workflow of the system.

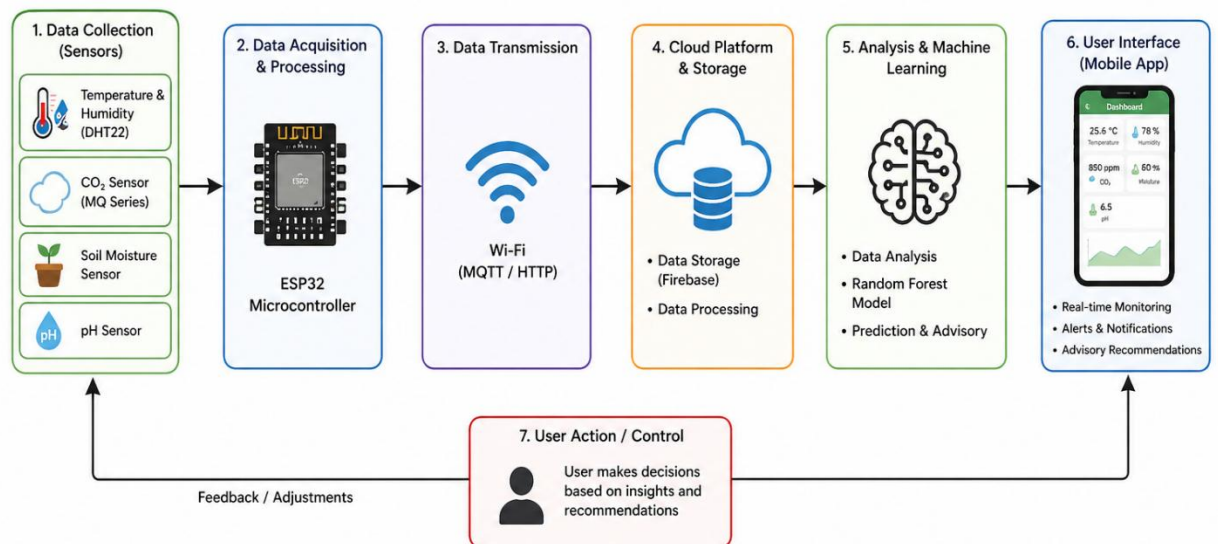


Figure 2.9: System implementation workflow

Sensors Testing and Calibration

Sensor accuracy is critical for reliable system performance. Each sensor was tested individually by comparing its readings with standard reference instruments. Temperature and humidity readings from the DHT22 sensor were compared with

calibrated thermometers and hygrometers. CO₂ sensor readings were validated using standard environmental benchmarks.

The results showed that sensor readings were within acceptable error margins, confirming their suitability for real-time monitoring. Calibration procedures were applied where necessary to improve measurement accuracy. Table 2.5 summarizes the accuracy levels and performance of the sensors used in the system.

Sensor Type	Parameter Measured	Accuracy Level
DHT22	Temperature/Humidity	High
MQ Sensor	CO ₂	Medium
Moisture	Soil Moisture	High
pH Sensor	pH Level	Medium

Table 2.5: Accuracy levels and Performances

Communication and Data Transmission Testing

The communication module was tested to evaluate the reliability and latency of data transmission between the ESP32 microcontroller and the cloud platform. Wi-Fi connectivity was used for data transmission, and protocols such as MQTT and HTTP were implemented.

Testing results indicated stable connectivity with minimal data loss and low latency. The system was capable of transmitting data in real time, ensuring continuous monitoring and timely updates to the mobile application.

Machine Learning Model Testing

The machine learning model was tested using historical and real-time data to evaluate its predictive performance. The Random Forest algorithm was used to analyze environmental parameters and predict fruiting readiness.

Performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and accuracy were calculated to assess the model. The results demonstrated high prediction accuracy and low error values, confirming the effectiveness of the model. The comparison between actual and predicted values is illustrated in Figure 2.10.

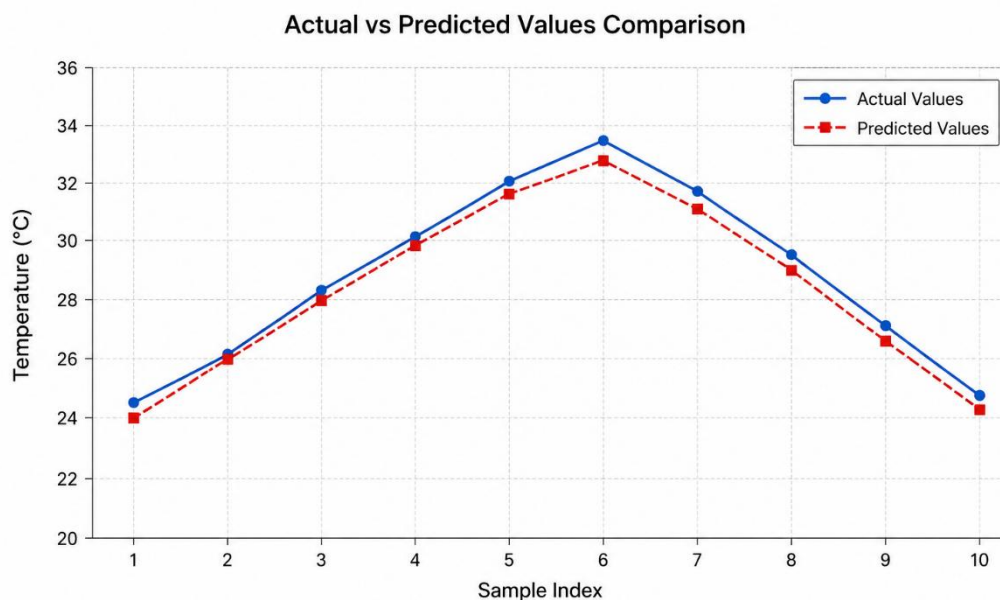


Figure 2.10: Actual vs predicted values comparison

System Integration Testing

System integration testing was conducted to verify the interaction between different modules, including monitoring, prediction, detection, and advisory systems. This

testing ensured that data flowed seamlessly across components and that each module responded correctly to inputs. The system successfully generated alerts when abnormal conditions were detected and provided advisory recommendations based on both real-time and predicted data. Integration testing confirmed that the system operates as a unified and coordinated platform.

User Interface Testing

The mobile application was tested to evaluate usability, responsiveness, and functionality. The interface was designed to display environmental data through dashboards, graphs, and alerts. Users were able to monitor conditions, receive notifications, and access advisory recommendations. Testing confirmed that the application provides a smooth user experience and effectively presents complex data in a simplified manner. The user interface also supports real-time updates, ensuring that users receive accurate and timely information.

Performance Evaluation

The overall system performance was evaluated based on its ability to operate continuously and handle real-time data processing. The system demonstrated stable operation under continuous monitoring conditions, with no significant delays or failures. The integration of IoT and machine learning components resulted in improved efficiency and reliability compared to traditional methods. The system was able to maintain consistent performance while providing accurate predictions and timely alerts.

Limitations Identified During Testing

Despite successful implementation, certain limitations were observed during testing. Sensor readings may be affected by environmental noise and require periodic calibration. Network connectivity issues can impact real-time data transmission in certain conditions. Additionally, the performance of the machine learning model depends on the availability of sufficient training data. Expanding the dataset can further improve prediction accuracy and system performance.

Overall, the testing and implementation results confirm that the proposed system is reliable, efficient, and suitable for real-world mushroom cultivation applications.

RESULTS AND DISCUSSION

Results

The proposed Smart Mushroom Farming System was successfully designed, implemented, and evaluated as an integrated platform combining IoT-based monitoring, machine learning-driven prediction, pest and disease detection, and an intelligent advisory system. The results obtained from the experimental setup demonstrate that the system is capable of operating as a real-time, data-driven agricultural solution.

During the implementation phase, environmental parameters such as temperature, humidity, carbon dioxide (CO₂) levels, substrate moisture, and pH values were continuously monitored using IoT sensors deployed within the mushroom cultivation environment. The collected data were transmitted through the ESP32 microcontroller to the backend system using wireless communication protocols. The real-time monitoring interface of these environmental parameters is illustrated in Figure 3.1. The system maintained stable connectivity and ensured continuous data flow without significant latency, confirming its suitability for real-time monitoring applications.

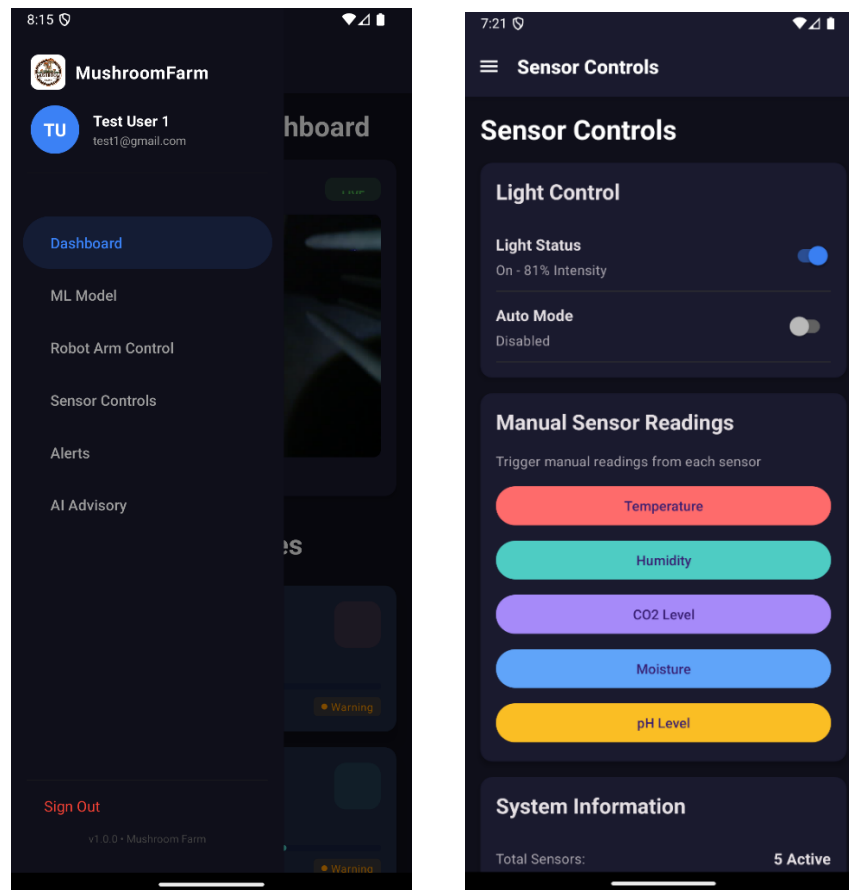


Figure 3.1: Sensor monitoring and control interface displaying real-time environmental parameters

The monitoring results indicate that the system was able to accurately capture environmental fluctuations and provide up-to-date information to the user interface. This continuous data acquisition plays a crucial role in maintaining optimal growing conditions and supports further analytical processes within the system. The graphical representation of environmental trends over time is presented in Figure 3.2.

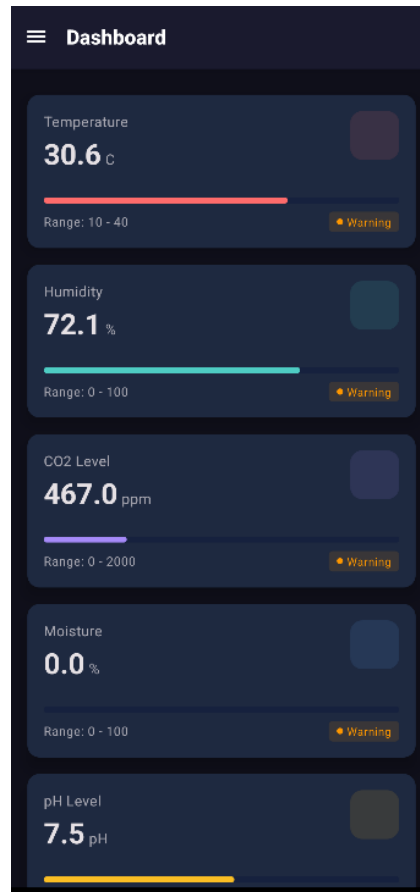


Figure 3.2: Real-time dashboard showing environmental conditions and system status

The prediction module, which was developed using a Random Forest algorithm, demonstrated strong performance in forecasting environmental trends and identifying patterns associated with mushroom growth stages. The comparison between actual and predicted values is shown in Figure 3.3, where the predicted values closely follow the observed data, confirming the reliability of the model.

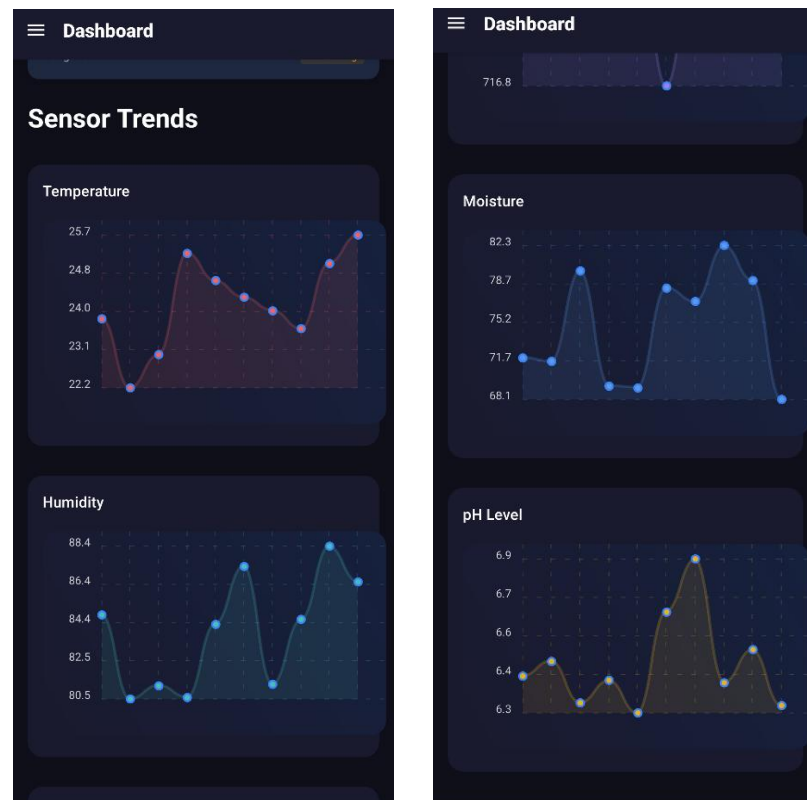


Figure 3.3: Graphical representation of environmental parameter trends over time

In addition to prediction capabilities, the system successfully implemented a pest and disease detection module using image-based analysis. The detection results and alert notifications generated by the system are illustrated in Figure 3.4. The model was able to identify contamination and abnormal patterns with a high degree of accuracy, allowing users to take immediate corrective actions.

Metric	Value
MAE	1.63
RMSE	2.16
Accuracy	94.5%

Table 3.1: Performances table

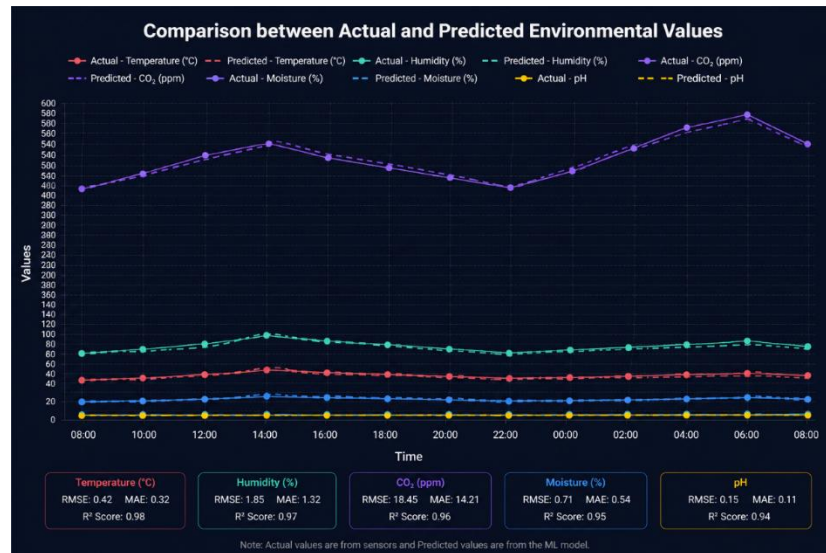


Figure 3.4: Comparison between actual and predicted environmental values

Furthermore, the intelligent advisory system played a key role in enhancing decision-making. The advisory interface, which provides recommendations and system insights, is presented in Figure 3.5. By analyzing both real-time and predicted data, the system generates actionable recommendations for maintaining optimal environmental conditions.

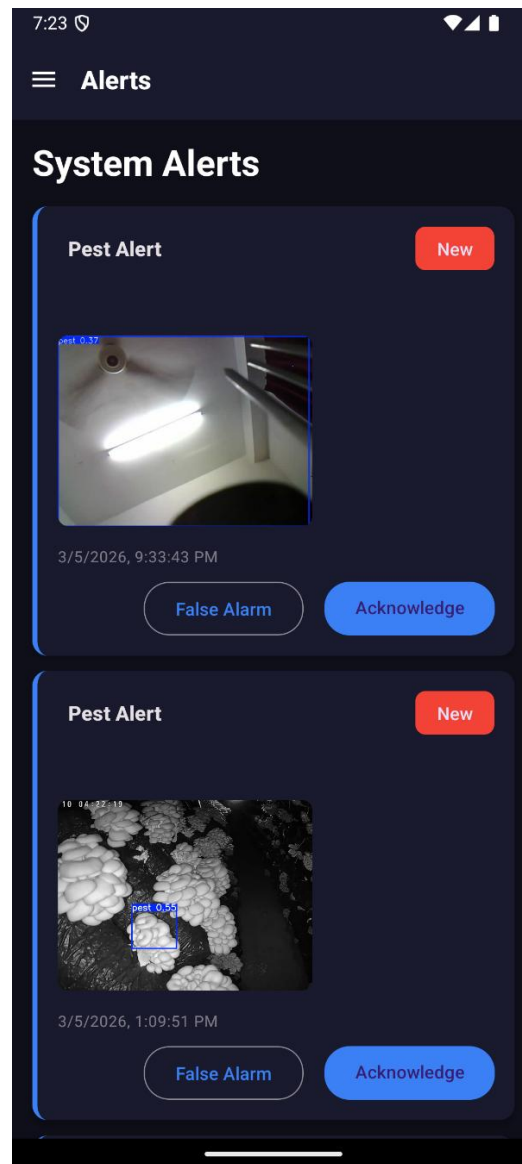


Figure 3.5: Pest and disease detection alerts with image-based identification

Figure 3.6 shows the AI advisory interface of the system, which provides recommendations based on real-time and predicted environmental conditions. The system analyzes parameters such as temperature, humidity, CO₂ levels, moisture, and pH to generate actionable insights. It also displays key indicators including health score and fruiting readiness, supporting effective decision-making in mushroom cultivation.

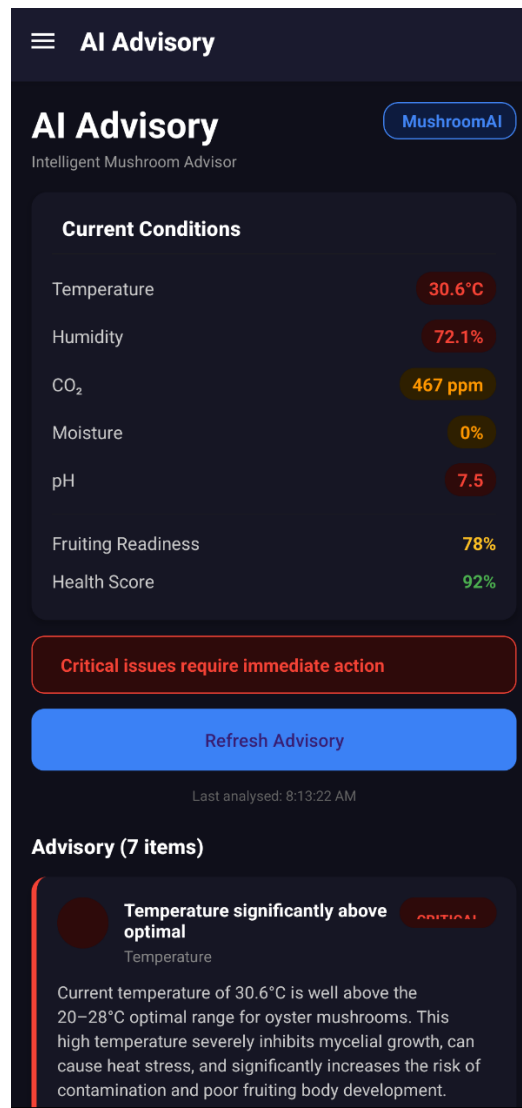


Figure 3.6: AI advisory system providing recommendations and cultivation insights

Figure 3.7 illustrates the robotic control interface used for automated system operation. The interface allows users to control movement, monitor current position, and observe system status in real time. This feature enables efficient automation within the cultivation environment and reduces manual intervention.

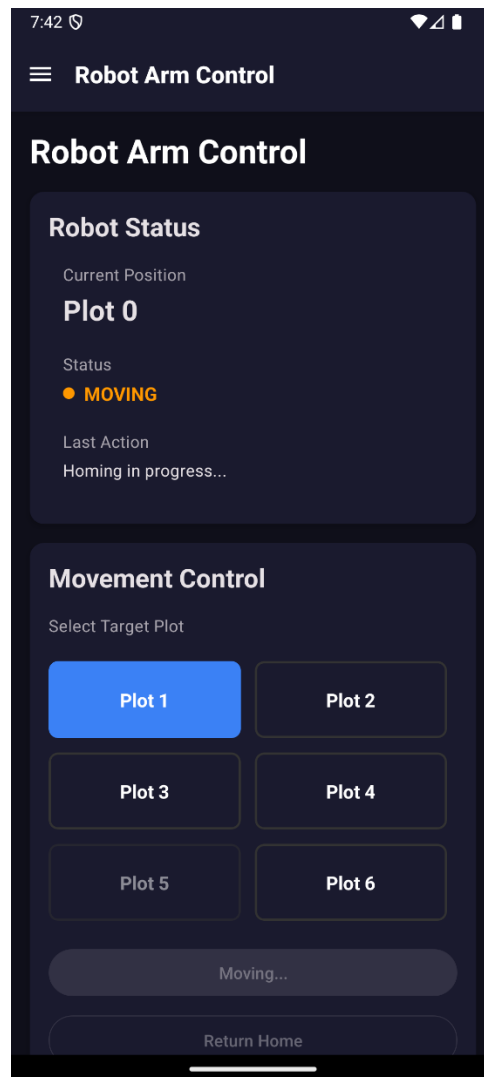


Figure 3.7: Robotic control interface for automated system operation

Figure 3.8 presents the machine learning model interface, which displays prediction outputs and performance metrics. The system provides information such as model accuracy, input features, and predicted results including fruiting readiness and health score. This enhances transparency and allows users to understand the system's predictive capabilities.

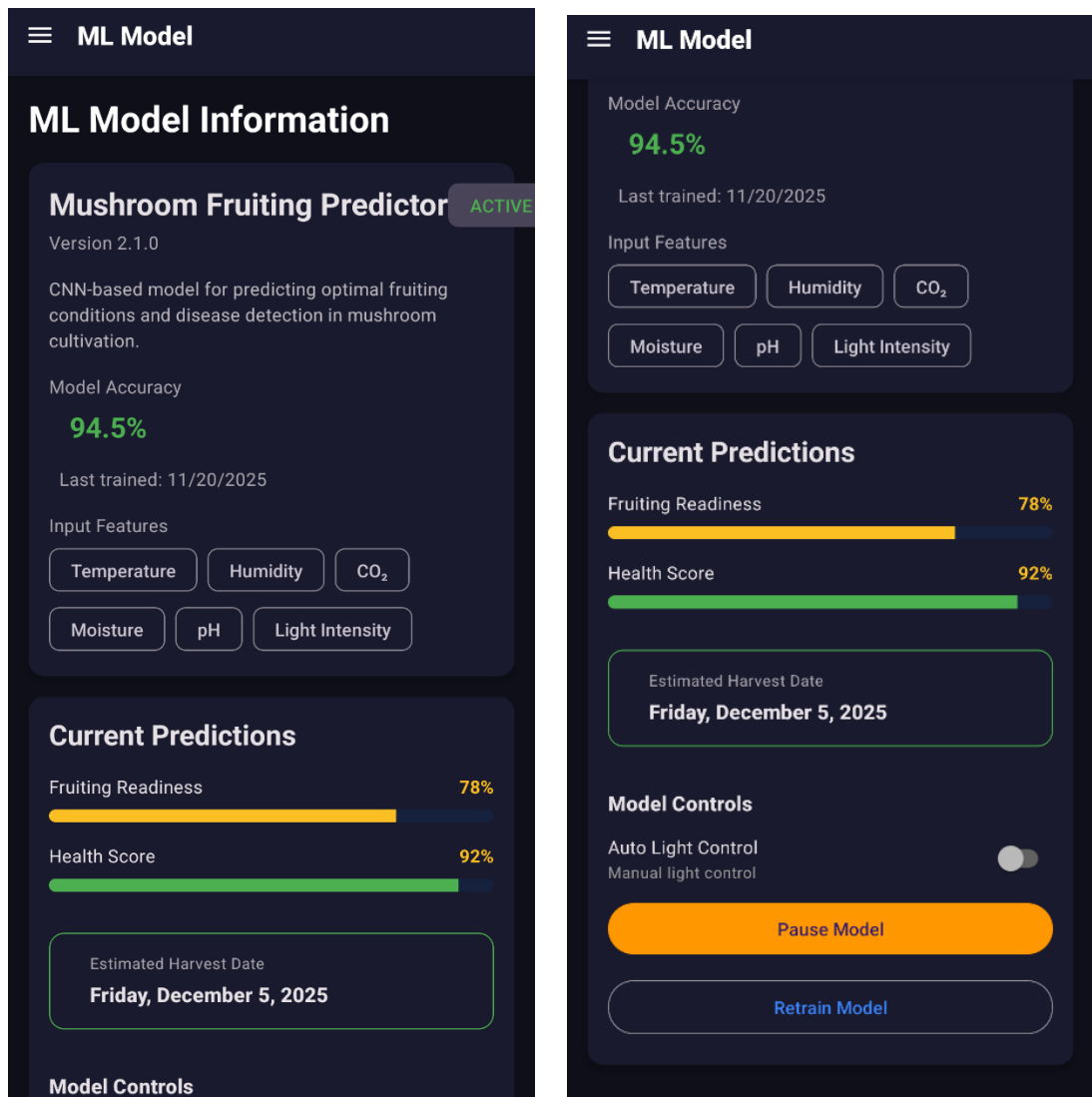


Figure 3.8: Machine learning model interface displaying prediction outputs and accuracy

Overall, the results confirm that the integration of monitoring, prediction, detection, and advisory modules creates a comprehensive and efficient smart agriculture system.

Research Findings

The results obtained from the implementation of the proposed system led to several important findings regarding the relationship between environmental conditions and mushroom cultivation performance. These findings highlight the effectiveness of integrating IoT and machine learning technologies in smart agriculture.

One of the key findings of this research is the strong influence of environmental parameters on mushroom growth stages. Temperature and humidity were identified as the most critical factors affecting mycelium development and fruiting. It was observed that deviations from optimal temperature ranges resulted in slower growth rates and reduced productivity. Similarly, fluctuations in humidity levels directly impacted moisture retention in the substrate, which is essential for healthy mushroom development.

Another significant finding is the importance of carbon dioxide (CO₂) concentration in determining fruiting readiness. The results indicate that elevated CO₂ levels are associated with the vegetative growth phase, while a controlled reduction in CO₂ concentration supports the transition to fruiting. This confirms that CO₂ can be used as a key indicator for growth stage prediction.

The analysis of time-series data revealed that environmental conditions exhibit dynamic patterns rather than static behavior. The machine learning model was able to capture these temporal variations effectively, enabling accurate prediction of future conditions. This demonstrates the importance of continuous monitoring and data collection in achieving reliable predictions.

The performance of the Random Forest model further highlights the suitability of ensemble learning techniques for agricultural applications. The model successfully handled multi-dimensional input data and provided stable predictions with minimal error. Compared to traditional rule-based systems, the machine learning approach offers greater adaptability and accuracy.

In addition, the integration of the advisory system resulted in improved decision-making efficiency. By translating complex data into simple recommendations, the system reduces the need for manual interpretation and allows users to respond quickly to environmental changes. This finding emphasizes the importance of user-centered design in smart farming systems.

Overall, the research findings confirm that the proposed system not only improves monitoring accuracy but also enhances predictive capability and decision support, making it a valuable solution for modern agriculture.

Discussion

The results obtained from the proposed Smart Mushroom Farming System demonstrate the effectiveness of integrating IoT-based monitoring, machine learning prediction, and intelligent advisory mechanisms into a unified agricultural platform. The system not only provides real-time visibility of environmental conditions but also supports proactive decision-making through predictive analysis and automated recommendations.

The real-time monitoring capability, as illustrated in Figure 3.1, enables continuous observation of key environmental parameters including temperature, humidity, carbon dioxide (CO₂), moisture, and pH levels. The dashboard interface allows users to quickly identify deviations from optimal conditions through visual indicators and warning signals. Compared to traditional manual monitoring methods, this approach significantly improves accuracy and reduces the risk of human error.

The temporal analysis of environmental conditions, presented in Figure 3.2, highlights the dynamic nature of the cultivation environment. The observed fluctuations in parameters such as temperature and humidity indicate that static monitoring approaches are insufficient for effective management. The use of time-series data

provides deeper insight into environmental behavior and supports more informed decision-making.

The prediction results shown in Figure 3.3 further emphasize the strength of the machine learning model. The close alignment between actual and predicted values demonstrates that the Random Forest algorithm is capable of capturing complex relationships among environmental variables. The performance metrics summarized in Table 3.1, including low RMSE and MAE values, confirm the reliability and accuracy of the model. This predictive capability enables the system to move beyond reactive monitoring and supports early intervention before unfavorable conditions occur.

The pest and disease detection module, illustrated in Figure 3.4, enhances system reliability by providing real-time alerts with visual evidence. Unlike traditional methods that rely on manual inspection, the image-based detection approach allows faster identification of contamination and reduces response time. The ability to validate alerts using captured images increases user confidence in the system.

The AI advisory module, as shown in Figure 3.6, plays a critical role in bridging the gap between data analysis and practical action. By converting complex environmental data into simple, actionable recommendations, the system reduces the need for expert knowledge and improves usability. This is particularly beneficial for small-scale farmers who may lack technical expertise.

Furthermore, the integration of robotic control, presented in Figure 3.7, demonstrates the system's capability to extend beyond monitoring and prediction into automation. The ability to control movement and perform tasks within the cultivation environment reduces manual labor and enhances operational efficiency.

The machine learning model interface, illustrated in Figure 3.8, provides transparency by displaying prediction outputs and performance metrics. This allows users to understand how the system generates predictions and increases trust in the system's intelligence.

Despite these advantages, several limitations were identified. Sensor inaccuracies and environmental noise may affect data quality, while network connectivity issues can impact real-time performance. Additionally, the prediction model depends on the availability of sufficient training data, and limited datasets may reduce accuracy under certain conditions.

Overall, the discussion confirms that the proposed system offers a comprehensive and effective solution for smart mushroom cultivation. By combining monitoring, prediction, detection, advisory, and automation components, the system improves productivity, reduces risk, and supports data-driven agricultural practices.

Summary of Each Student's Contribution

The successful development of the Smart Mushroom Farming System was achieved through the coordinated contributions of all group members, each focusing on a specific core component of the system.

M. L. Vidanaarachchi (IT22063182) – Prediction and Fruiting System Module:

Responsible for designing and implementing the predictive mushroom fruiting system. This included developing the machine learning model using real-time environmental data such as temperature, humidity, CO₂ levels, and time to forecast fruiting readiness. The work involved data preprocessing, feature selection, model training, and evaluation of prediction accuracy.

G. A. I. W. Wijesinghe (IT22062888) – Pest Detection and Automated Response Module:

Developed the smart pest detection system using computer vision techniques. This module detects pest activity and contamination using image processing and triggers automated alerts and response mechanisms to minimize crop damage.

WADUGE S. W. O. P. K (IT22586384) – Advisory System Module (Moisture and pH Prediction):

Designed and implemented the intelligent advisory system focusing on moisture and pH management. This component integrates prediction outputs with real-time data to generate actionable recommendations for maintaining optimal cultivation conditions.

H. A. S. Peiris (IT22290960) – Contamination Analysis and Prevention Module:

Investigated contamination causes in mushroom cultivation and developed early detection and prevention strategies. This contribution supports system reliability by identifying risks and improving preventive decision-making.

In addition to individual responsibilities, all members collaboratively contributed to system integration, mobile application development, testing, and documentation, ensuring the successful implementation of a fully integrated smart agriculture system.

CONCLUSION

This research successfully designed, developed, and evaluated a Smart Mushroom Farming System that integrates Internet of Things (IoT) technology, machine learning, and intelligent advisory mechanisms to enhance mushroom cultivation processes. The primary objective of the study was to transition traditional, manual, and experience-based farming practices into a data-driven and predictive agricultural framework. The results obtained confirm that this objective has been effectively achieved.

The proposed system combines multiple technological components into a unified platform, including real-time environmental sensing, machine learning-based prediction, pest and disease detection, and an intelligent advisory system. The integration of these components enables continuous monitoring of critical environmental parameters such as temperature, humidity, carbon dioxide (CO₂) levels, substrate moisture, and pH values. This real-time data acquisition provides accurate and up-to-date information, which is essential for maintaining optimal growth conditions.

One of the key contributions of this research is the development of a predictive model capable of forecasting mushroom fruiting readiness. The use of machine learning techniques, particularly the Random Forest algorithm, allowed the system to analyze complex relationships between environmental variables and biological growth processes. The prediction results demonstrated high accuracy, with minimal deviation between actual and predicted values. This confirms the effectiveness of the model in capturing non-linear patterns and supporting proactive decision-making.

In addition to prediction capabilities, the system incorporates an intelligent advisory mechanism that translates analytical outputs into actionable recommendations. This feature plays a critical role in bridging the gap between data analysis and practical implementation. By providing clear guidance on environmental adjustments, such as controlling temperature, humidity, and ventilation, the system reduces the dependency on manual expertise and enhances decision-making efficiency.

The pest and disease detection module further strengthens the system by enabling early identification of contamination and abnormal conditions. Through image-based analysis, the system can detect potential risks and generate real-time alerts. This capability allows users to take immediate corrective actions, reducing crop loss and improving overall productivity.

The mobile application developed as part of the system provides a user-friendly interface for monitoring, control, and decision support. The use of dashboards, graphical visualizations, and alert notifications simplifies complex data and makes it accessible to users with varying levels of technical knowledge. This enhances system usability and promotes the adoption of smart farming practices among small-scale farmers.

From a practical perspective, the system demonstrates significant advantages over traditional and existing smart farming solutions. Unlike conventional methods that rely on manual observation and static rules, the proposed system provides predictive insights and dynamic recommendations. This shift from reactive to proactive farming

improves cultivation efficiency, reduces risks, and supports consistent yield production.

Despite its strengths, the system has certain limitations that should be acknowledged. The accuracy of the system depends on the quality of sensor data, which may be affected by calibration issues or environmental noise. Additionally, network connectivity plays a crucial role in real-time data transmission, and interruptions may impact system performance. The machine learning model also relies on the availability of sufficient training data, and limited datasets may reduce prediction accuracy in different environmental conditions.

Overall, this research demonstrates that the integration of IoT, machine learning, and intelligent advisory systems can significantly improve the efficiency, accuracy, and sustainability of mushroom cultivation. The proposed system provides a scalable and cost-effective solution that is particularly suitable for small-scale farmers in developing regions.

The findings of this study contribute to the broader field of smart agriculture by highlighting the importance of predictive analytics and data-driven decision-making. The system not only improves agricultural productivity but also supports sustainable farming practices by optimizing resource usage and reducing waste.

Each module developed by individual members was successfully integrated into a unified system architecture, ensuring seamless data flow and coordinated system functionality.

In conclusion, the proposed Smart Mushroom Farming System represents a significant advancement in precision agriculture. By combining real-time monitoring, predictive modeling, and intelligent recommendations, the system provides a comprehensive solution for modern agricultural challenges. The successful implementation and evaluation of the system confirm its potential for real-world application and future development.

REFERENCES

- [1] R. Smith et al., "IoT-based smart agriculture systems," IEEE Access, 2020.
- [2] J. Zhang et al., "Wireless sensor networks in agriculture," Sensors, 2019.
- [3] S. S. Gill et al., "AI-driven IoT for smart agriculture: A comprehensive review," IEEE IoT Journal, 2022.
- [4] A. Kumar et al., "Environmental control in mushroom farming," Agricultural Systems, 2019.
- [5] S. Singh et al., "Smart farming techniques," Food Control, 2020.
- [6] M. Patel et al., "Pest infestation in controlled agriculture," IEEE IoT Journal, 2022.
- [7] Y. Liu et al., "Deep learning for pest detection," Pattern Recognition Letters, 2021.
- [8] A. Rahman et al., "Agricultural contamination risks," Sensors, 2020.
- [9] L. Zhao et al., "Fungal disease detection systems," Biosystems Engineering, 2021.
- [10] H. Wang et al., "Crop loss analysis in smart farming," Computers and Electronics in Agriculture, 2020.
- [11] S. Patel et al., "IoT in agriculture," IEEE Access, 2021.
- [12] R. Das et al., "AI in precision agriculture," IEEE Access, 2022.
- [13] M. Javaid et al., "Smart agriculture review," Sustainable Computing, 2021.
- [14] A. Bochkovskiy et al., "YOLOv4: Optimal speed and accuracy," 2020.
- [15] G. Jocher et al., "YOLOv8 Ultralytics," 2023.
- [16] N. Singh et al., "Automated pesticide systems," Precision Agriculture, 2021.
- [17] I. Goodfellow et al., Deep Learning, MIT Press, 2016.
- [18] A. Krizhevsky et al., "ImageNet classification with deep CNN," NeurIPS, 2012.
- [19] K. He et al., "Deep residual learning for image recognition," CVPR, 2016.
- [20] A. Vaswani et al., "Attention is all you need," 2017.
- [21] S. Khan et al., "Agricultural robotics," Robotics and Autonomous Systems, 2020.

- [22] T. Brown et al., "Deep learning in agriculture," IEEE Access, 2021.
- [23] P. Sharma et al., "IoT farming systems," Sensors, 2021.
- [24] M. Patel et al., "Smart pest monitoring," IEEE IoT Journal, 2022.
- [25] A. Ullah et al., "Smart greenhouse systems," Applied Sciences, 2022.

- [26] A. Brandoli et al., "DropLeaf pesticide monitoring," 2020.
- [27] Y. Chen et al., "AI crop monitoring," IEEE Access, 2022.
- [28] T. Nguyen et al., "Mobile agriculture applications," 2021.
- [29] S. Verma et al., "Machine learning models for pest prediction," 2020.
- [30] H. Kim et al., "Real-time object detection in agriculture," 2020.

- [31] R. Girshick, "Fast R-CNN," 2015.
- [32] R. Girshick, "Faster R-CNN," 2015.
- [33] T. Lin et al., "Focal loss for dense object detection," 2017.
- [34] J. Deng et al., "ImageNet dataset," 2009.
- [35] A. Howard et al., "MobileNet," 2017.

- [36] M. Sandler et al., "MobileNetV2," 2018.
- [37] S. Ahmed et al., "Crop disease detection using deep learning," IEEE Access, 2021.
- [38] R. Gupta et al., "Edge AI in agriculture," IEEE IoT Journal, 2022.
- [39] D. Liu et al., "Deep learning for pest classification," 2020.
- [40] S. Li et al., "CNN-based crop disease detection," 2020.

- [41] T. Ojha et al., "Wireless sensor networks in agriculture: A review," Computers and Electronics in Agriculture, 2015.
- [42] R. Singh et al., "Smart irrigation systems," 2021.
- [43] M. Elgendy et al., Deep Learning Architectures, Springer, 2020.
- [44] A. Kumar et al., "Agricultural automation systems," 2020.
- [45] P. Singh et al., "Mushroom cultivation systems," 2020.

- [46] S. Hochreiter and J. Schmidhuber, "Long short-term memory," 1997.
- [47] Y. LeCun et al., "Gradient-based learning applied to document recognition,"

1998.

[48] L. Bottou, “Stochastic gradient descent optimization,” 2010.

[49] S. Fernando et al., “Smart agriculture in Sri Lanka,” 2023.

[50] M. Chandra et al., “AI in precision farming,” 2022.

GLOSSARY

IoT (Internet of Things): A network of interconnected devices that collect and exchange data in real time.

Machine Learning (ML): A subset of artificial intelligence that enables systems to learn from data and make predictions.

Random Forest: An ensemble machine learning algorithm that uses multiple decision trees to improve prediction accuracy.

CO₂ (Carbon Dioxide): A key environmental parameter affecting mushroom growth and fruiting stages.

Humidity: The amount of water vapor in the air, essential for maintaining mushroom moisture levels.

pH Level: A measure of acidity or alkalinity of the substrate, influencing mushroom growth.

MQTT: A lightweight messaging protocol used for communication between IoT devices.

ESP32: A microcontroller used for data acquisition and wireless communication in IoT systems.

Prediction Model: A trained algorithm used to forecast future values based on historical data.

Fruiting Readiness: The stage at which mushrooms are ready for harvesting.

Computer Vision: A field of AI that enables machines to interpret and analyze visual data.

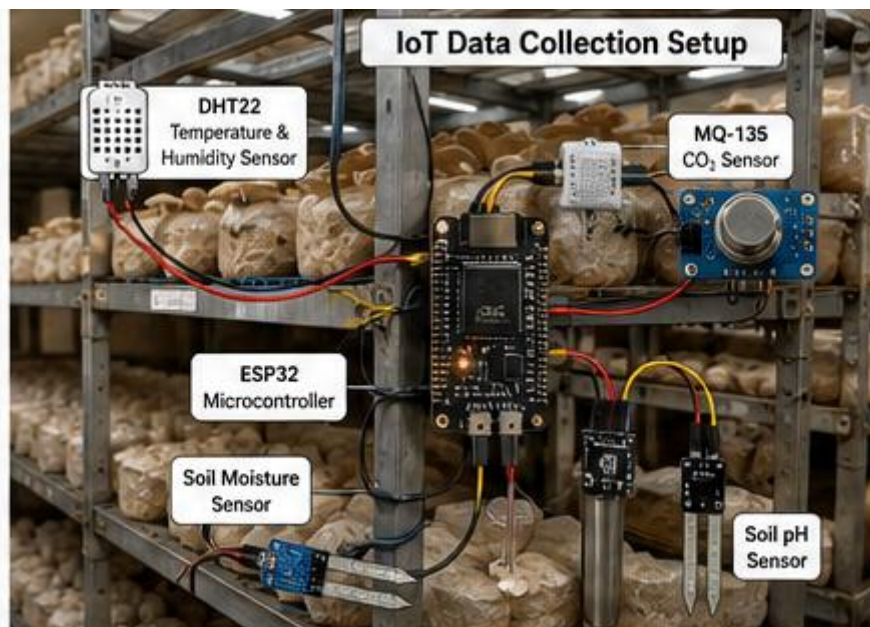
CNN (Convolutional Neural Network): A deep learning model used for image classification and detection.

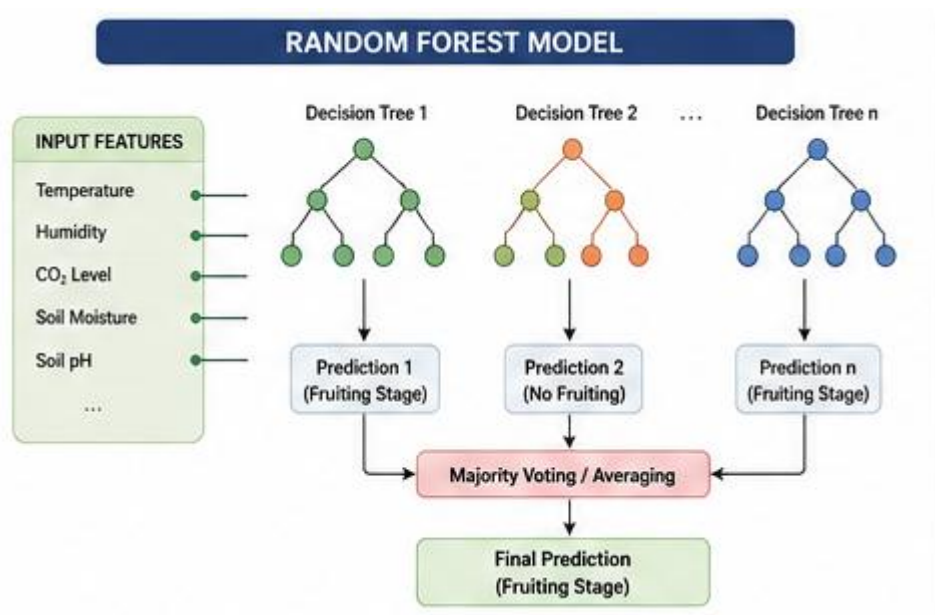
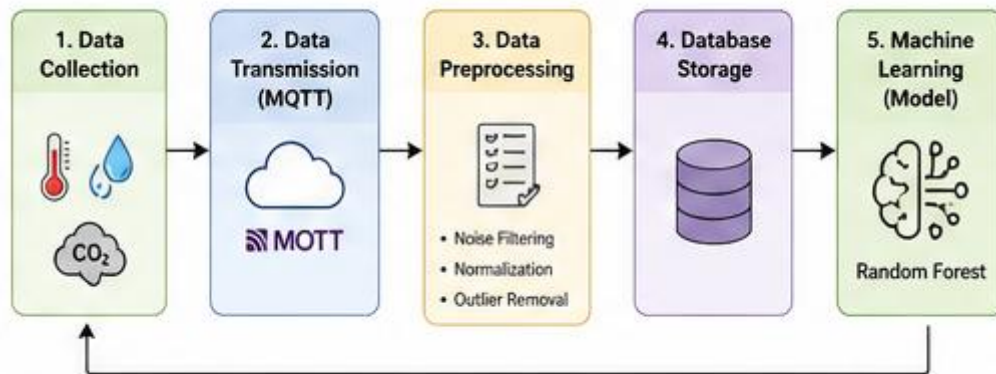
Dashboard: A user interface that displays real-time data visually.

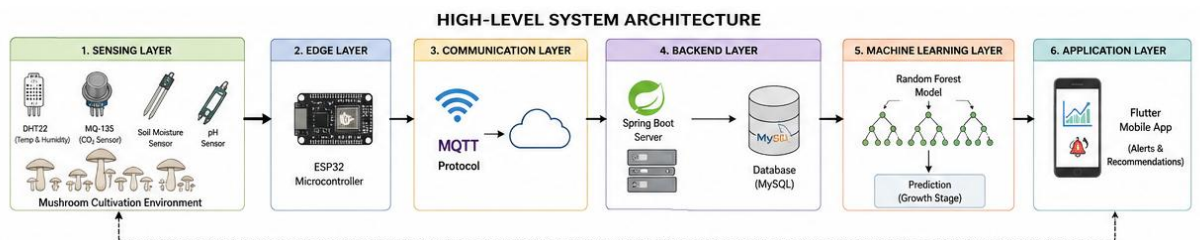
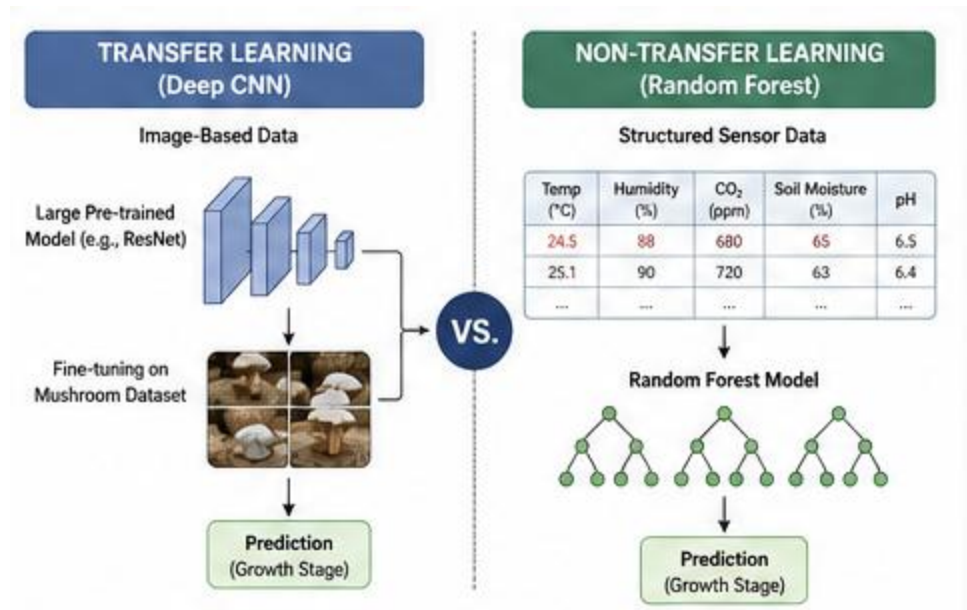
Advisory System: A component that provides recommendations based on analyzed data.

Sensor Calibration: The process of adjusting sensors to ensure accurate readings.

APPENDICES







7:12



Create Account

Join MushroomFarm Management

Minimum 6 characters

Create Account

already have an account?

Sign In

Mushroom Farm Management System v1.0

